Chow Varieties

25110840012 上海数学中心 谢景炎

Contents

1 Introduction

The study of parameter spaces for algebraic cycles is a foundational cornerstone of algebraic geometry. While the Hilbert scheme successfully parameterizes closed subschemes of a projective variety by keeping track of the scheme-theoretic structure, there is frequently a need to parameterize uniruled or effective algebraic cycles where the nilpotent structure is forgotten and only the formal sum of subvarieties and their multiplicities is retained. The parameter space that achieves this is called **Chow variety**.

This note explores the representability of the Chow functor, tracing its classical roots to its modern functorial formulation. We will outline the construction and representability in characteristic 0 via the Chow form, and subsequently analyze the geometric and algebraic pathologies that arise in characteristic $p > 0$, highlighting why classical approaches fail and how the moduli problem must be adjusted.

1.1 Historical context

The concept of parameterizing varieties of a given dimension and degree dates back to the classical era of algebraic geometry. The foundations were laid in 1937 by W. L. Chow and B. L. van der Waerden in their seminal paper *Zur algebraischen Geometrie IX*, where they introduced the **Chow form** (or often called the Cayley form) to assign a set of homogeneous coordinates to any projective variety.

By associating an algebraic cycle to a specific point in a projective space, they demonstrated that the set of all effective cycles of a fixed dimension and degree forms a closed algebraic set—the Chow variety. In the 1950s and 60s, Grothendieck incorporated the Chow variety into the modern scheme-theoretic framework, notably establishing the **Hilbert-Chow morphism**, which provides a natural map from the Hilbert scheme of closed subschemes to the Chow variety of cycles. However, defining the Chow variety as representing a well-behaved functor of *families of cycles* over arbitrary base schemes remained a delicate problem, leading to extensive modern treatments by mathematicians such as D. Mumford, J. Kollár, and B. Rydh.

1.2 Representability in characteristic 0

In characteristic 0, the representability of the Chow variety can be established robustly using the classical theory of Chow forms and symmetric products. The main steps to construct the Chow variety $\mathrm{Chow}_{d,k}(\mathbb{P}^n_S)$ (parameterizing cycles of dimension k and degree d in $\mathbb{P}^n_S$) and prove its representability through following main steps:

- **Step 1: Construction of the** Ch **Map.** We begin by defining a natural transformation from the functor of families of cycles to a projective space. For a given family of cycles, we construct the Chow map (the Ch morphism), which assigns to each cycle its corresponding point in a projective space $\mathbb{P}(V)$. This projective space parameterizes effective Cartier divisors of a specific degree on the relative Grassmannian.

- **Step 2: The Chow Form.** The precise algebraic object underlying the Ch map is the Chow form. For an individual irreducible k-dimensional subvariety $X \subset \mathbb{P}^n_S$, we consider the incidence correspondence between X and the Grassmannian $\mathbb{G}(n-k-1, n)$. The locus of $(n-k-1)$-planes that intersect X forms a hypersurface in the Grassmannian. The single defining section of this hypersurface is the Chow form F_X. This construction extends additively to formal cycles.

- **Step 3: Construction of the Universal Family.** To show representability, we must construct a universal family of cycles over the closed subscheme defined by the image of the Ch map. This involves constructing a closed subscheme $\mathcal{U} \subset \text{Chow} \times \mathbb{P}^n_S$ that universally parameterizes the cycles, derived from the universal properties of the incidence variety and the zero-locus of the universal Chow form.

- **Step 4: Completing the Proof.** Finally, we prove that the Ch map factors through a well-defined closed subscheme (the Chow variety) and that pulling back the universal family $\mathcal{U}$ via this map recovers the original family of cycles. In characteristic 0, the complete reducibility of symmetric groups allows the symmetric quotients to behave perfectly, ensuring that this correspondence is both bijective and functorial, thereby completing the proof of representability.

1.3 Obstructions in positive characteristic

When the characteristic of the base field is $p > 0$, the elegant correspondence between algebraic cycles and their Chow forms breaks down, introducing severe obstructions to representability.

- **Purely Inseparable Extensions:** The primary obstruction arises from purely inseparable field extensions. If an algebraic cycle is covered by a family where the generic fiber is geometrically non-reduced (which can happen in characteristic p due to purely inseparable morphisms), the classical Chow form fails to accurately capture the multiplicity of the cycle.

- **Failure of Symmetric Powers:** The functorial correspondence relies heavily on symmetric products to define additions of cycles. In characteristic p, taking the geometric quotient of a scheme by the symmetric group S_d behaves poorly. The map from an affine space to its symmetric product is no longer separable when p divides d, meaning the symmetric tensors no longer accurately represent the "sets of unordered points" in families.

- **Topological vs. Scheme-Theoretic Fibers:** Because the Chow form can evaluate to a p^e-th power for inseparable families, the classical definition only determines the cycle up to a Frobenius twist. The natural Hilbert-Chow morphism Hilb $\to$ Chow may have purely inseparable fibers, meaning the Chow variety, as classically constructed, is functionally closer to a topological space than a space carrying the correct scheme-theoretic moduli structure.

To resolve these obstructions in modern algebraic geometry, the functorial definition of families of cycles must be modified. Instead of relying on symmetric powers, modern proofs (such as those by Rydh) utilize **divided powers** (or operations in the ring of Witt vectors) to keep track of multiplicities correctly, allowing for a well-behaved moduli stack or scheme even in mixed and positive characteristics.

2 Family of Cycles

2.1 Algebraic cycles

Recall some basic definitions and properties for algebraic cycles first.

Definition 2.1. *Let X be of finite type over S. A d-dimensional algebraic cycle on X/S is a formal linear combination $\sum_i a_i[V_i]$, where V_i are irreducible closed subschemes of X over a 0-dimension closed subscheme of S. Denote $Z_d(X)$ to be the set of all d-dimensional algebraic cycles on X/S. We say a cycle is effective if $a_i \geq 0$ for all i. In particular, the fundamental cycle $[X] := \sum_i m_i[X_i]$, where X_i are d-dimensional irreducible components of X and $m_i = \mathrm{length}(\mathcal{O}_{X,X_i})$.*

For a finite S-morphism $f : X \to Y$, define pushforward $f_* : Z_d(X) \to Z_d(Y)$ sending $[V]$ to $\deg(V/W)[W]$, where W is the schematic closure of $f(V)$ and $\deg(V/W) = [K(V) : K(W)]$ the extension degree of function fields.

For a flat S-morphism $f : X \to Y$ of relative dimension r, define pullback $f^* : Z_d(Y) \to Z_{d+r}(X)$ sending $[W]$ to $[f^{-1}(W)]$, where $f^{-1}(W)$ is identified with $X \times_Y W$. Note that if f is surjective, then f^* is injective.

Let $Z = \sum_i a_i[V_i]$ be a d-dimensional algebraic cycle on X/S, D Cartier divisor on X. Define the degree of Z with respect to D to be $\deg_D Z = \sum_i a_i(V_i \cdot D^d)$, where $V_i \cdot D^d$ is the intersection product.

Assume X is over k. Then for any field extension K/k, $p : X_K := X \times_k K \to X$ is faithfully flat of relative dimension 0. This gives an injective pullback map $p^* : Z_d(X) \to Z_d(X_K)$. For $Z \in Z_d(X_K)$, we say Z defined over a subfield $k \subset F \subset K$ if Z lies in the image of $Z_d(X_F) \to Z_d(X_K)$.

Remark 2.1. *For convenience, in many references, we would also call formal linear combination $\sum_i a_i[V_i]$ cycle even if V_i are not of same dimension. Here we call them quasi-cycles to distinguish from cycles.*

Lemma 2.1. *Given a fibered diagram*

$$\begin{array}{ccc} X' & \xrightarrow{f'} & Y' \\ \downarrow{\scriptstyle g'} & & \downarrow{\scriptstyle g} \\ X & \xrightarrow{f} & Y \end{array} \tag{1}$$

where f is finite and g is flat. Then for any $Z \in Z_d(X)$, $g^* \circ f_*(Z) = f'_* \circ g'^*(Z)$..

Proposition 2.1. *Let X be a scheme, $Z = \sum_i a_i[V_i] \in Z_d(X)$. There is a unique closed subscheme W of X such that the fundamental cycle $[W] = Z$ if one of the following conditions holds*

- *$a_i = 1$ for all i.*
- *X is nonsingular at codimension 1 and has pure dimension $d+1$.*

Lemma 2.2. *Let X be a scheme over field k, D Cartier divisor on X. If K/k is a field extension, $X_K := X \times_k K$ with projection $p : X_K \to X$, then for all $Z \in Z_d(X)$, $\deg_{p^*D} p^*Z = \deg_D Z$.*

Proof. First note that $p^*(Z \cdot D^d) = p^*Z \cdot (p^*D)^d$ in Chow group. Suffice to show for all closed point $x \in X$ and $f^*[x] = \sum_i m_i[y_i]$, we have that $\chi(X,x) = \sum_i m_i \chi(X_K, y_i)$. In fact, this immediately comes from the equation

$$[k(x) : k] = \sum_i \mathrm{length}(\mathcal{O}_{X_K \times_X k(x), y_i}) \cdot [k(y_i) : K] \tag{2}$$

since $k(x) \otimes_k K \cong \prod_i \mathcal{O}_{X_K \times_X k(x), y_i}$. □

Lemma 2.3. *Let X be a scheme over field k of characteristic 0 and K/k field extension, $Z \in Z_d(X_K)$. Then there is a minimal field $k \subset k' \subset K$ such that X is defined over k'.*

Definition 2.2. *Let X be a scheme over field k, K/k and L/k field extensions. We say two cycles $Z \in Z_d(X_K)$ and $Z' \in Z_d(X_L)$ are essentially equivalent if for any field F with embeddings $K \hookrightarrow F$ and $L \hookrightarrow F$, $Z_F = Z'_F$, denoted by $Z \overset{ess}{=} Z'$. Equivalently, $Z \overset{ess}{=} Z'$ if there is a purely inseparable extension k'/k such that $Z_{K^{p^{-\infty}}}$ and $Z'_{L^{p^{-\infty}}}$ are both defined over k' by same cycle, where the subscript $p^{-\infty}$ denote the perfect closure.*

Remark 2.2. *When characteristic 0, $Z \overset{ess}{=} Z'$ if and only if they are bothe defined over k by same cycle.*

By definition, we immediately get the following corollaries.

Corollary 2.1. *$Z \overset{ess}{=} Z'$ if and only if $Z_{K^{p^{-\infty}}} \overset{ess}{=} Z_{L^{p^{-\infty}}}$.*

Corollary 2.2. *Notation as above, if $Z \overset{ess}{=} Z'$, then for any Cartier divisor D on X, $\deg_D Z = \deg_D Z'$.*

Remark 2.3. *Hence two essentially same cycles have nearly same numerical property.*

2.2 Family of cycles

Lemma 2.4. *Let $T = \operatorname{Spec} R$ be spectrum of DVR with generic point T_g and closed point T_0. For any morphism $f : X \to T$, there is closed subscheme $F_T(X)$ of X flat over T such that for any closed subscheme $Y \subseteq X$ flat over T, we have $F_T(X) \subseteq Y$. We call X'_T the flat closure of generic fiber.*

Definition 2.3. *Let $g : V \to W$ be a proper morphism, V irreducible and W reduced. By generic flatness, there is an open dense subset $W_0 \subset W$ such that $g^{-1}(W_0) \to W_0$ is flat of relative dimension d. Given a morphism $h : T \to W$ from $T = \operatorname{Spec} R$ spectrum of DVR sending T_0 to w and T_g to $w_g \in W_0$, consider the fibered product $V_T := V \times_W T$. Take the flat closure $F_T(V_T)$ of generic fiber, then $F_T(U_T)$ is flat over T of relative dimension d. In fact, its special fiber is of pure dimension d. Set*

$$\lim_{h \to w} (V/W) := [F_T(U_T) \times_T T_0] \in Z_d(g^{-1}(w) \times_T T_0) \tag{3}$$

called the cycle-theoretic fiber of g at w at h.

Remark 2.4. *To see the special fiber is of pure dimension d, take irreducible component V' of $F_T(V_T)$. By Going-down Theorem, generic point of V' would map to T_g. While g is proper, we get V' dominates T. Then by Krull's Main Theorem, we see special fiber is of dimension d.*

Lemma 2.5. *The definition of cycle-theoretic fiber is independent on the choice of W_0.*

Definition 2.4. *Let $g : V \to W$ be a proper morphism between reduced schemes. Assume $C(V) = \sum_i m_i[V_i]$ is quasi-cycle over V, where V_i varies over all irreducible components of V with $m_i \neq 0$. Denote $g_i = g\big|_{V_i}$. We say $(g : C(V) \to W)$ is a quasi-well defined family of d-dimensional algebraic cycles over W if*

- *V_i maps onto some irreducible component of W for all i.*
- *Every fiber of g_i is either empty or of dimension d. In particular there is an open dense subset $W_0 \subset W$ such that $g_i^{-1}(W_0) \to W_0$ is flat of relative dimension d.*

and if moreover it satisfies

- *(Condition $\star$) For all $w \in W$, there is a cycle $g^{[-1]}(w) \in Z_d(g^{-1}(w) \times_{k(w)} k(w)^{p^{-\infty}})$ such that*

$$g^{[-1]}(w) \overset{ess}{=} \sum_i m_i \lim_{h \to w} (V_i/W) \tag{4}$$

for all $h : T \to W$ from $T = \operatorname{Spec} R$ spectrum of DVR sending T_g into W_0.

then we say $(g : C(V) \to W)$ is a well defined family of d-dimensional algebraic cycles over W.

Remark 2.5. *The condition $\star$ seems rather complicated. But we will see that if W is normal and $(g : C(V) \to)$ quasi-well defined, then condition $\star$ holds automatically. Also, when characteristic 0, we see $g^{[-1]}(w) \in Z_d(g^{-1}(w))$, which is more natural.*

Definition 2.5. *Let X be a scheme over S, W reduced scheme over S. A well defined family of d-dimensional algebraic cycles of X/S over W/S is a well defined family of d-dimensional algebraic cycles $(g : C(V) \to W)$ such that $V \subseteq X \times_S W$ is a closed subscheme and $g : V \to W$ is the natural projection.*

Proposition 2.2. *Let X be a scheme over S, D Cartier divisor on X. Assume $(g : C(V) \to W)$ be a well defined family of d-dimensional cycles of X/S over W/S and W is connected. Then $\deg_D g^{[-1]}(w)$ is constant.*

With inspiration, we would give a definition of degree of a well defined family of cycles.

Definition 2.6. *Let X be a scheme projective over S, $\mathcal{O}_X(1)$ given by the projective embedding. For $(g : C(V) \to W)$ be a well defined family of d-dimensional cycles of X/S over W/S, if $\deg_{\mathcal{O}_X(1)} g^{[-1]}(w)$ is constant, then we call the constant d' the degree of $(g : C(U) \to W)$.*

Remark 2.6. *Hence by Proposition 2.2, when W is connected, well defined family $(g : C(V) \to W)$ has degree.*

Proposition 2.3. *Let $(g : C(V) \to W)$ be quasi-well defined family. Then $(g : C(V) \to W)$ is well defined family if and only if there are commutative diagrams*

$$\begin{tikzcd} V'_j \arrow[r] \arrow[d, "\widetilde{g}_j"] & \widetilde{V} \arrow[d, "\widetilde{g}"] \\ W' \arrow[r, "p"] & W \end{tikzcd} \tag{5}$$

where $\widetilde{V}$ is a scheme over W such that $\widetilde{V}^{\text{red}} = V$, $p : W' \to W$ is a proper, surjective morphism and $V'_j \subset \widetilde{V} \times_W W'$ closed subschemes flat over W' for all j, and there are integers m'_j such that

- *for all $w \in W_0$ and $w' \in p^{-1}(w)$, we have*
$$\sum_j m'_j[\widetilde{g}_j^{-1}(w')] \overset{ess}{=} \sum_i m_i[g_i^{-1}(w)] \tag{6}$$
- *for all $w \in W$ and $w'_1, w'_2 \in p^{-1}(w)$, we have*
$$\sum_j m'_j[\widetilde{g}_j^{-1}(w'_1)] \overset{ess}{=} \sum_j m'_j[\widetilde{g}_j^{-1}(w'_2)] \tag{7}$$

Corollary 2.3. *Let $g : V \to W$ be a proper morphism between reduced schemes, flat of relative dimension d. Assume V is of pure dimension and take $C(V) = [V]$ the fundamental cycle of V. Then $(g : C(V) \to W)$ is a well defined family of cycles.*

Corollary 2.4. *Let $(g : C(V) \to W)$ be a well defined family of cycles. Assume $h : T \to W$ is a morphism from spectrum of DVR sending T_0 to w and T_g into W_0. Then*

$$g^{-1}(w) = \operatorname{Supp} g^{[-1]}(w) = \cup_i \operatorname{Supp}(\lim_{h \to w}(V_i/W)) \text{ as sets} \tag{8}$$

where all these supports refer to their image under projection to $g^{-1}(w)$ and hence the equation holds as subspaces of $g^{-1}(w)$.

Lemma 2.6. *Let X be a scheme over field k, $(g : C(V) \to W)$ a well defined family of cycles of X/k over W/k. Assume W is geometrically connected. Then $g^{[-1]}(w)$ is essentially independent of $w \in W$ i.e. for any $w, w' \in W$, we have*

$$g^{[-1]}(w) \overset{ess}{=} g^{[-1]}(w') \tag{9}$$

Theorem 2.1. *Let $(g : C(V) \to W)$ be a quasi-well defined family of cycles, W normal. Then $(g : C(V) \to W)$ is in fact well defined family.*

2.3 Chow pullback

Let $(g : C(V) \to W)$ be a well defined family of cycles. Assume $f : W' \to W$ is a morphism with W' reduced, satisfying that $g^{[-1]}(f(w'))$ is defined over $k(f(w'))$ for all $w' \in W'$. Then one can define a well defined family of cycles over W' as the following lemma.

Lemma 2.7. *Notations as above. Take V'_j to be irreducible components of $\widetilde{V} := V \times_W W'$ with reduced scheme structure. Set $V'_{j,w'}$ the fiber of $V'_j \to W'$ at generic point $w' \in W'$. Assume w' maps to w and $g^{[-1]}(w) = \sum_k a_k[Z_k]$, where Z_k vary over all d-dimensional irreducible closed subsets of $g^{-1}(w)$. Then*

- *Z_k are all irreducible components of $g^{-1}(w)$ and $[Z_k \times_w w'] = \sum_j b_{k,j}[V'_{j,w'}]$ for some integers $b_{k,j}$.*

Define $m'_j := \sum_{w' \text{ generic}} \sum_k a_k b_{k,j}$. Set $f^{\mathrm{ch}}V := \cup_{j \text{ for } m'_j \neq 0} V'_j$ with reduced scheme structure, $f^{\mathrm{ch}}g : f^{\mathrm{ch}}V \hookrightarrow \widetilde{V} \xrightarrow{\widetilde{g}} W'$ and $f^{\mathrm{ch}}C(V) = C(f^{\mathrm{ch}}V) := \sum_j m'_j[V'_j]$. Then

- *$(f^{\mathrm{ch}}g : C(f^{\mathrm{ch}}V) \to W')$ is a well defined family of d-dimensional cycles over W'.*

called the Chow pullback of $(g : C(V) \to W)$ along f. We may also denote it by $f^{\mathrm{ch}}(g : C(V) \to W)$.

Proof. By definition, $\dim g^{-1}(w) = d$ so that Z_k are irreducible components. By Corollary 2.4, we know $g^{-1}(w)$ is of pure dimension d so that Z_k vary over all irreducible components of $g^{-1}(w)$. Also, since w' is generic point, $V'_{j,w'}$ would be either empty or an irreducible component of $\widetilde{g}^{-1}(w')$. For those irreducible $V'_{j,w'}$, its image under $\widetilde{g}^{-1}(w') \to g^{-1}(w)$ is also irreducible hence contained in some Z_k. Note that by some useful result, $Z_k \times_w w'$ is of pure dimension d, we get that $V'_{j,w'}$ is of dimension d for all j and generic point $w' \in W'$ so that $\dim \widetilde{g}^{-1}(w')$ is purely d-dimensional. Thus $[Z_k \times_w w'] = \sum_j b_{k,j}[V'_{j,w'}]$ for some integers $b_{k,j}$.

By upper semicontinuity, we get $\widetilde{g}^{-1}(w'') \geq d$ for all $w'' \in W'$. While $\widetilde{g}^{-1}(w'') = \cup_j V'_j \times_{W'} w''$ as sets and $V'_j \times_{W'} w''$ is contained in $g^{-1}(f(w'')) \times_{g(w'')} w''$ for some k, then $\dim \widetilde{g}^{-1}(w'') = \sup_j \dim(V'_j \times_{W'} w'') \leq \dim(g^{-1}(f(w'')) \times_{f(w'')} w'') = d$. Thus for all $w'' \in W'$, we get $\dim \widetilde{g}^{-1}(w'') = d$. Similarly, we also get $\dim(f^{\mathrm{ch}}g)^{-1}(w'') = d$. As properness is stable under base change and composition, we get $f^{ch}g$ is proper. Hence $(f^{\mathrm{ch}}g : C(f^{\mathrm{ch}}V) \to W')$ is a quasi-well defined family of d-dimensional cycles.

□

Lemma 2.8. *Let $(g : C(V) \to W)$ be a well defined family, $f_1 : W_1 \to W$ and $f_2 : W_2 \to W_1$ morphisms with W_1 and W_w reduced. Then*

$$f_2^{\mathrm{ch}} \circ f_1^{\mathrm{ch}}(g : C(V) \to W) \cong (f_1 \circ f_2)^{\mathrm{ch}}(g : C(V) \to W) \tag{10}$$

3 Seminormal and Weakly Normal

Definition 3.1. *Let $f : X \to Y$ be a finite surjective morphism between reduced schemes. Define the seminormalization (resp. weak normalization) of Y with respect to f to be a scheme $\widetilde{Y}_f^{\mathrm{semi}}$ (resp. $\widetilde{Y}_f^{\mathrm{weak}}$) together with a finite surjective morphism $\widetilde{f} : \widetilde{Y} \to Y$ such that*

- *f factors through $\widetilde{f}$*
- *$\widetilde{f}$ is bijective*
- *for all $y' \in \widetilde{Y}_f^{\mathrm{semi}}$, $k(\widetilde{f}(y')) \to k(y')$ is isomorphic.*
- *(resp. for all $y' \in \widetilde{Y}_f^{\mathrm{weak}}$, $k(\widetilde{f}(y')) \to k(y')$ is purely inseparable extension and if $\widetilde{f}(y)$ is generic point, then isomorphic)*
- *if $f' : Y' \to Y$ satisfies properties above, then $\widetilde{f}$ uniquely factors through f'*

For excellent reduced Y, the normalization map $\widetilde{Y} \to Y$ is finite surjection. Then we define the seminormalization (resp. weak normalization) of Y to be the seminormalization (resp. weak normalization) of Y with respect to $\widetilde{Y} \to Y$, denoted by $\widetilde{Y}^{\mathrm{semi}}$ (resp. $\widetilde{Y}^{\mathrm{weak}}$). If $Y \cong \widetilde{Y}^{\mathrm{semi}}$ (resp. $Y \cong \widetilde{Y}^{\mathrm{weak}}$), then we say Y is seminormal (resp. weakly normal).

Remark 3.1. *Clearly, seminormal implies weakly normal and when characteristic 0, seminormalization is same to weak normalization.*

From now on, schemes would be assumed to be excellent.

Lemma 3.1. *Let $f : X \to Y$ be a finite surjective morphism between reduced schemes. Then $\widetilde{Y}_f^{\mathrm{semi}}$ and $\widetilde{Y}_f^{\mathrm{weak}}$ are both reduced.*

Proof. Here we only proves for seminormalization, same argument can be applied to weak normalization. Since the reduction map is a bijective closed immersion hence finite surjective morphism, by universal property of reduction, we get $\mathrm{red}\, \widetilde{Y}_f^{\mathrm{semi}} \to Y$ also satisfies the conditions. Hence by universal property of seminormalization, $\widetilde{Y}_f^{\mathrm{semi}} \to Y$ uniquely factors through $\mathrm{red}\, \widetilde{Y}_f^{\mathrm{semi}} \to Y$. Thus there is a commutative diagram

$$\begin{tikzcd} \mathrm{red}\, \widetilde{Y}_f^{\mathrm{semi}} \arrow[rr, shift left, "i"] \arrow[dr] & & \widetilde{Y}_f^{\mathrm{semi}} \arrow[ll, shift left, "j"] \arrow[dl] \\ & Y & \end{tikzcd} \tag{11}$$

Then by universal property of reduction and seminormalization, we get $j \circ i = \mathrm{id}$ and $i \circ j = \mathrm{id}$. Conclude that $\widetilde{Y}_f^{\mathrm{semi}} \cong \mathrm{red}\, \widetilde{Y}_f^{\mathrm{semi}}$ and hence reduced. $\square$

Lemma 3.2. *Let $f : Y \to X$ be a morphism between reduced schemes, Y seminormal. Then f uniquely factors as $Y \to \widetilde{X}^{\mathrm{semi}} \to X$.*

Let $X \to S$ be a smooth and projective morphism of relative dimension N, $(g : C(V) \to W)$ well defined family of $(N-1)$-dimensional cycles of X/S over W/S. Assume w_g is a generic point of W. Since $X \times_S W \times_W k(w_g)^{p^{-\infty}}$ is smooth, by Proposition 2.1, $g^{[-1]}(w_g)$ is fundamental cycle of some Cartier divisor $D(w_g)$ on $X \times_S W \times_W k(w_g)^{p^{-\infty}}$. As $D(w_g)$ is flat over $k(w_g)^{p^{-\infty}}$, it corresponds to a morphism $\operatorname{Spec} k(w_g) \to \mathrm{CDiv}(X \times_S W/W)$. Take the schematic closure W' of morphism $k(w_g) \to \mathrm{CDiv}(X \times_S W/W)$. And $p : W' \to W$ the projection.

Lemma 3.3. *Notations as above, we have that*

- *W' is dominated by the weak normalization of W.*
- *Assume W' is seminormal. Then p is isomorphic if and only if $g^{[-1]}(w_g)$ is defined over $k(w_g)$.*
- *If W is weakly normal, then g is flat.*

4 Chow Varieties

Now it's time for us to define Chow functor and prove its representability when characteristic 0.

Definition 4.1. *Let X be a scheme over S. For all reduced scheme W over S, define*

$$\mathcal{C}how(X/S)(W) := \{\text{well defined family of effective cycles of } X/S \text{ over } W/S\} \tag{12}$$

If X moreover is projective over S with line bundle $\mathcal{O}_X(1)$, define

$$\mathcal{C}how_{d,d'}(X/S)(W) := \{(g : C(V) \to W) \in \mathcal{C}how(X/S)(W) \text{ of dimension } d \text{ and degree } d'\} \tag{13}$$

Hence by Proposition 2.2, for connected W, we have

$$\mathcal{C}how(X/S)(W) = \sqcup_{d,d'} \mathcal{C}how_{d,d'}(X/S)(W) \tag{14}$$

Theorem 4.1. *Notations as above. There is contravariant functor*

$$\mathcal{C}how_{d,d'}(X/S)(\cdot) : \{\text{seminormal } S\text{-scheme}\} \longrightarrow \mathbf{Set} \tag{15}$$

And the functor is represented by a seminormal scheme $\mathrm{Chow}_{d,d'}(X/S)$ which is projective over S, with universal family $\mathrm{Univ}_{d,d'}(X/S)$.

Functoriality immediately comes from the results in Chow pullback. We will prove the representability step by step. To be clarified, during the proof, we would not assume characteristic 0 so that it is clearer to see the role of characteristic 0.

Step 1: Construction of Chow map

In this step, for $\mathbb{P}_S$ the projective bundle over S associated to some rank $n+1$ vector bundle, we construct a natural transfromation from $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)$ to a Chow functor parameterizing hypersurfaces.

Let $\mathbb{P}_S^\vee$ be the dual bundle. Consider the incidence correspondence

$$I := \{(x, H_1, \cdots, H_{d+1}) \in \mathbb{P}_S \times_S (\mathbb{P}_S^\vee)^{d+1} | x \in H_1 \cap \cdots \cap H_{d+1}\} \tag{16}$$

Denote $p_1 : I \to \mathbb{P}_S$ and $p_2 : I \to (\mathbb{P}_S^\vee)^{d+1}$ the projections. Notice that p_1 and p_2 are both smooth morphisms.

Given $s : W \to S$ and well defined family $(g : C(V) \to W)$ of $\mathbb{P}_S/S$ over W/S, there is a diagram

$$\begin{tikzcd} V \arrow[r, hook] & s^*\mathbb{P}_S & s^*I \arrow[l, "s^*p_1"'] \arrow[d, "s^*p_2"] \\ & & s^*(\mathbb{P}_S^\vee)^{d+1} \end{tikzcd} \tag{17}$$

For quasi-cycle Z on V, set $\mathrm{Ch}(Z) := (s^*p_2)_* \circ (s^*p_1)^*(Z)$. Also define $\mathrm{Ch}(V_i) = \mathrm{red}(s^*p_2)_* \circ (s^*p_1)^*(V_i)$.

Lemma 4.1. *The projection $p : (s^*p_1)^*V_i \to \mathrm{Ch}(V_i)$ is generically finite and generically injective i.e. for an open dense subset U of $\mathrm{Ch}(V_1)$, the morphism $p^{-1}(U) \to U$ is finite and injective. In particular, when characteristic 0, we get $\mathrm{Ch}([V_i]) = [\mathrm{Ch}(V_i)]$.*

Proof. □

Lemma 4.2. *For $W = \operatorname{Spec} k$ spectrum of a field, the projection $p : (s^*p_1)^*V_i \to \mathrm{Ch}(V_i)$ is a local isomorphism at $(x, H_1, \cdots, H_{d+1})$ if and only if*

$$V_i \cap H_1 \cap \cdots \cap H_{d+1} = \{x\} \tag{18}$$

as schematic intersection.

Proposition 4.1. *Notations as above, then we have*

- $(\mathrm{Ch}(g) : \mathrm{Ch}(C(V)) \to W)$ *is a well defined family of $(dn + n - 1)$-dimensional cycles of $(\mathbb{P}_S^\vee)^{d+1}/S$ over W/S, for convenience, we may this is a well defined family of hypersurfaces.*
- $\mathrm{Ch}(C(V)) = \sum_i m_i k_i [\mathrm{Ch}(V_i)]$ *for some integers k_i. If every geometric generic fiber of g is reduced, then $k_i = 1$ for all i. In particular, this holds if characteristic 0.*
- *Let $f : W' \to W$ be a morphism between reduced schemes. Then*

$$(\mathrm{Ch}(f^{\mathrm{ch}} g) : \mathrm{Ch}(f^{\mathrm{ch}} C(V)) \to W') \cong (f^{\mathrm{ch}} \mathrm{Ch}(g) : f^{\mathrm{ch}} \mathrm{Ch}(C(V)) \to W') \tag{19}$$

- *There is a unique closed subscheme $\mathrm{Ch}(V) \subset (\mathbb{P}_S^\vee)^{d+1}$ such that $\mathrm{Ch}(V)$ is purely one-codimensional, has no embedded point and its fundamental cycle is $\mathrm{Ch}(C(V))$.*

Step 2: Chow forms

Definition 4.2. *Let K/k be a field extension, $Z \in Z_d(\mathbb{P}_K)$. Then $\mathrm{Ch}(Z)$ is hypersurface of $(\mathbb{P}_K^\vee)^{d+1}$ defined by a multihomogeneous polynomial $\mathrm{ch}(Z)$, called the Chow form or Cayley form of cycle Z. In addition, the minimal field of definition $\mathrm{Ch}(Z)$ over k is called the chow field of Z, denoted by $k^{\mathrm{ch}}(Z)$.*

Lemma 4.3. *Let $Z{=}\sum_i a_i[V_i] \in Z_d(\mathbb{P}_K)$. Then*

- *If $E \supset K$ is a field extension, then $\mathrm{ch}(Z_E) = \mathrm{ch}(Z_K)$.*
- *There is a one-to-one correspondence*

$$\{\textit{components of } Z\} \leftrightsquigarrow \{\textit{irreducible divisor of } \mathrm{ch}(Z)\} \tag{20}$$

  In fact, we have $\mathrm{ch}(Z) = \prod_i (\mathrm{ch}([V_i]))^{a_i}$.

- *Set $d' = \deg_{\mathcal{O}(1)}$, then $\mathrm{ch}(Z)$ is of multidegree $(d', \cdots, d')$.*

Corollary 4.1. *Assume S is spectrum of a field of characteristic 0. Denote $q : (\mathbb{P}_S^\vee)^{d+1}$ the projection and $\mathbb{H}_S := \mathrm{Proj}_S(q_* \mathcal{O}_{(\mathbb{P}_S^\vee)^{d+1}}(d', d', \cdots, d'))$, which parameterizes hypersurfaces of multidegree $(d', d', \cdots, d')$ in $(\mathbb{P}_S^\vee)^{d+1}$. Then there is a natural transformation*

$$\mathrm{Ch} : \mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathbb{H}) \tag{21}$$

Proof. By Proposition 4.1, we sends $(g : C(V) \to W)$ to a hypersurface $\mathrm{Ch}(V)$, which corresponds to a morphism $W \to \mathrm{Hilb}((\mathbb{P}_S^\vee)^{d+1}/S)$. And by Lemma 4.3, we know the chow form $\mathrm{ch}(C(V))$ is of multidegree $(d', \cdots, d')$. Hence the morphism factors through $\mathbb{H}$. □

We also define inverse map of Ch.

Definition 4.3. *First set-theoretically define*

$$\mathrm{Ch}^{-1}(H) := \{x \in \mathbb{P}_S \big| p_1^{-1}(x) \subset p_2^{-1}(H)\} \tag{22}$$

for hypersurface $H \subset (\mathbb{P}_S^\vee)^{d+1}$.

Proposition 4.2. *For all hypersurface $H \subset (\mathbb{P}_S^\vee)^{d+1}$, $\dim(\mathrm{Ch}^{-1}(H)) \leq d$. And for irreducible and reduced H, if $\dim(\mathrm{Ch}^{-1}(H)) = d$, then $\mathrm{Ch}^{-1}(H)$ is irreducible of dimension d. And if moreover $S = \mathrm{Spec}\, k$ for some perfect field k, then $\mathrm{Ch}([\mathrm{Ch}^{-1}(H)]) = H$, where $[\mathrm{Ch}^{-1}(H)]$ is fundamental cycle of $\mathrm{Ch}^{-1}(H)$ with the reduced scheme structure.*

Corollary 4.2. *Assume k is perfect, then there is a one-to-one correspondence*

$$\{d\textit{-dimensional cycles of degree } d' \textit{ in } \mathbb{P}_k\} \leftrightsquigarrow \{H \in \mathbb{H}_k \big| \dim(\mathrm{Ch}^{-1}(H_j)) = d \textit{ for all irreducible components } H_j \textit{ o}$$

In particular, this holds when characteristic 0.

Proof. For H as above, assume $[H] = \sum_j a_j H_j$ and set $\mathrm{Ch}^{-1}([H]) = \sum_j a_j[\mathrm{Ch}^{-1}(H_j)]$. Check that Ch^{-1} is then inverse of *chowmap*. □

Step 3: Construction of universal family

As a preliminary step, we set

$$\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S) := \{H \in \mathbb{H}_S \big| \dim(\mathrm{Ch}^{-1}(H_j)) = d, \forall \text{ irreducible components } H_j \text{ of } H\}$$

Let $h : \mathbb{H}_S \to S$ be the projection map. Take universal family $\mathbb{U}_S$ of $\mathbb{H}_S$ which is a hypersurface of $(h^*\mathbb{P}_S^\vee)^{d+1}$. There is a diagram

$$\begin{tikzcd} h^*\mathbb{P}_S & h^*I \arrow[l, "h^*p_1"'] \arrow[d, "h^*p_2"] \\ & (h^*\mathbb{P}_S^\vee)^{d+1} \end{tikzcd} \tag{23}$$

By Proposition 4.2, we know $\dim(\mathrm{Ch}^{-1}(\mathbb{U}_S)) \leq d$. And there is a projection map $\pi : \mathrm{Ch}^{-1}(\mathbb{U}_S) \to \mathbb{H}_S$. Hence each topological fiber of π has dimension at most d.

Lemma 4.4. *We have an equivalent description of* $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *that*

$$\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S) = \{h \in \mathbb{H}_S \big| (\pi)^{-1}(H) \text{ has pure dimension } d\} \tag{24}$$

Proposition 4.3. $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *is a closed subset of* $\mathbb{H}_S$.

When S is spectrum of a field, Corollary 4.2 tells us that the morphism $W \to \mathbb{H}$ corresponding to $(g : C(V) \to W)$, given by Corollary 4.1, would again factor through $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$.

Now for general S, let W be a seminal scheme and $s : W \to S$ a morphism. Assume $(\widetilde{h} : C(\mathbf{H}) \to W)$ is a well defined family of hypersurfaces of $\mathbb{P}_S/S$ over W/S such that for any field F with morphism $\mathrm{Spec}\, F \to W$, the determining polynomial of hypersurface $\mathbf{H} \times_W F$ is Chow form of some cycle in $Z_d(\mathbb{P}_F)$.

In order to define $\mathrm{Ch}^{-1}(C(\mathbf{H}))$, we need to assign multiplicities of irreducible components.

Lemma 4.5. *Define* $\mathrm{Ch}^{-1}(C(\mathbf{H}))$ *as following. Then* $\mathrm{Ch}^{-1}(\widetilde{h} : C(\mathbf{H}) \to W) := (\mathrm{Ch}^{-1}(\widetilde{h}) : \mathrm{Ch}^{-1}(C(\mathbf{H})) \to W)$ *is a well defined family of cycles. In addition,* Ch^{-1} *is the inverse of Chow map* Ch.

Proof. To do this, we may localize at generic point of $w_g \in W$. By assumption, there is a unique cycle over $\mathbb{P}_{k(w_g)}$ with Chow form determining $\mathbf{H} \times_W k(w_g)$. This would assign the multiplicity. □

Proposition 4.4. *When characteristic* 0*, for* $f : W' \to W$ *morphism between seminormal schemes, we have that*

$$\mathrm{Ch}^{-1} \circ f^{\mathrm{ch}}(\widetilde{h} : C(\mathbf{H}) \to W) \cong f^{\mathrm{ch}} \circ \mathrm{Ch}^{-1}(\widetilde{h} : C(\mathbf{H}) \to W) \tag{25}$$

Definition 4.4. *Take* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ *to be the seminormalization of* $\mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$*. In particular,* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ *is seminormal and projective over* S*. Set*

$$\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S) := \mathrm{Ch}^{-1}(\mathbb{U}_S \times_{\mathbb{H}_S} \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$$

and

$$C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) := \mathrm{Ch}^{-1}(C(\mathbb{U}_S \times_{\mathbb{H}_S} \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)))$$

which together give a well defined family $(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$*, called universal well defined family of* $\mathbb{P}_S/S$*.*

Remark 4.1. *By Lemma 3.2, we know that a morphism* $W \to \mathrm{Chow}'_{d,d'}(\mathbb{P}_S/S)$ *from a seminormal scheme* W *would factor through* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$*. Hence we finally get a natural transformation from* $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot)$ *to* $\mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$*.*

Now for X projective over S, choose an embedding $X \hookrightarrow \mathbb{P}_S$. Let

$$\mathrm{Chow}'_{d,d'}(X/S) := \{z \in \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S) \big| u^{-1}(z) \subset X \times_S k(z)\}$$

Lemma 4.6. $\mathrm{Chow}'_{d,d'}(X/S)$ *is a closed subset of* $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$*.*

Take $\mathrm{Chow}_{d,d'}(X/S)$ to be the seminormalization of $\mathrm{Chow}'_{d,d'}(X/S)$. Define the universal well defined family $(u_X : C(\mathrm{Univ}_{d,d'}(X/S)) \to \mathrm{Chow}_{d,d'}(X/S))$ of X/S to be the Chow pullback of well defined family of $\mathbb{P}_S/S$ along $\mathrm{Chow}_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. Clearly, $\mathrm{Chow}_{d,d'}(X/S)$ is seminormal and projective over S.

Step 4: Complete the proof

Remains to use the universal family to show the representability when characteristic 0. First check that $\mathcal{C}how_{d,d'}(\mathbb{P}_S/S)$ is representable by $\mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. As we remarked above, there is a natural transformation

$$\varphi : \mathcal{C}how_{d,d'}(\mathbb{P}_S/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)) \tag{26}$$

Also for any seminormal W over S, by Lemma 4.5, $\varphi(W)$ is a bijection. It is clear that $(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$ is the universal family.

Now for X projective over S, choose embedding $X \hookrightarrow \mathbb{P}_S$. As $X \times_S W$ is a closed subscheme of $\mathbb{P}_S \times_S W$, for all well defined family $(g : C(V) \to W)$ of X/S over W/S, it is also a well defined family of $\mathbb{P}_S/S$ over W/S. Hence it corresponds to a morphism $f : W \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$ and

$$(g : C(V) \to W) = f^{\mathrm{ch}}(u : C(\mathrm{Univ}_{d,d'}(\mathbb{P}_S/S)) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)) \tag{27}$$

Then by construction of Chow pullback, we know for all generic point $w_g \in W$, either $u^{-1}(f(w_g)) = \varnothing$ or $u^{-1}(f(w_g)) \times_{f(w_g)} w_g = V \times_W w_g \subset X \times_S w_g$. While $u^{-1}(f(w_g)) \times_{f(w_g)} w_g$ surjectively maps to $u^{-1}(f(w_g))$, we get $f(w_g) \in \mathrm{Chow}'_{d,d'}(X/S)$. Thus f factors through

$\mathrm{Chow}'_{d,d'}(X/S)$. Again by Lemma 3.2, f factors through $\mathrm{Chow}_{d,d'}(X/S)$. Finally we get a natural transformation

$$\varphi' : \mathcal{C}how_{d,d'}(X/S)(\cdot) \longrightarrow \mathrm{Hom}(\cdot, \mathrm{Chow}_{d,d'}(X/S)) \tag{28}$$

Note that $\varphi'(W)$ is restriction of the map $\varphi(W)$ hence injective. For surjectivity, since each morphism $W \to \mathrm{Chow}_{d,d'}(X/S)$ also gives a morphism in $\mathrm{Hom}(W, \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S))$. Hence there is a well defined family $(g : C(V) \to W)$ of $\mathbb{P}_S/S$ over W/S, which is the Chow pullback of universal well defined family of $\mathbb{P}_S/S$ along $W \to \mathrm{Chow}_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$.

We may first pull back along $\mathrm{Chow}'_{d,d'}(X/S) \to \mathrm{Chow}_{d,d'}(\mathbb{P}_S/S)$. Denote the Chow pullback $(g' : C(V') \to \mathrm{Chow}'_{d,d'}(X/S))$. Then $V \subseteq V' \times_{\mathrm{Chow}'_{d,d'}(X/S)} W$. While by definition, $V' \subseteq X \times_S \mathrm{Chow}'_{d,d'}(X/S)$, we get $V \subseteq X \times_S W$. Thus $(g : C(V) \to W)$ is in fact a well defined family of X/S over W/S. Conclude that $\varphi'(W)$ is bijective so that $\mathcal{C}how_{d,d'}(X/S)(\cdot)$ is representable by $\mathrm{Chow}_{d,d'}(X/S)$. As $\mathrm{Univ}_{d,d'}(X/S)$ is the Chow pullback of $\mathrm{Univ}_{d,d'}(\mathbb{P}_S \times_S S)$, it is the universal family of φ'.